

EVA Telemetry Ranges

As outlined by the mission description, it is your job to alert a user of values outside of the ranges specified below. The alerts, whether audible or visual, can be designed as you see fit based on your hardware/software. We have included some additional details explaining how these telemetry values interact with each other, and actions to take during an error scenario.

Value Name	Units	Min	Max
primary_battery_level	%	20	100
secondary_battery_level	%	20	100
oxy_pri_storage	%	20	100
oxy_sec_storage	%	20	100
oxy_pri_pressure	psi	600	3,000
oxy_sec_pressure	psi	600	3,000
coolant_storage	%	80	100
heart_rate	bmp	50	160
oxy_consumption	psi/min	0.05	0.15
co2_production	psi/min	0.05	0.15
suit_pressure_oxy	psi	3.5	4.1
suit_pressure_co2	psi	0.0	0.1
suit_pressure_other	psi	0.0	0.5
suit_pressure_total	psi	3.5	4.5
helmet_pressure_co2	psi	0.0	0.15
fan_pri_rpm	rpm	29,000	31,000
fan_sec_rpm	rpm	29,000	31,000
scrubber_a_co2_storage	%	0	60
scrubber_b_co2_storage	%	0	60
temperature	°C	10	32
coolant_liquid_pressure	psi	100	700
coolant_gas_pressure	psi	0	700

Suit Resources:

These values are filled up while using the UIA in the Egress procedures of the EVA. The Battery is filled up when the DCU is connected to the UIA, and the Oxygen Tanks are filled up during the procedure. These values are dropping throughout the extent of the EVA. The user should be alerted if the values are nearing their minimum nominal value, even if they are still within range.

The coolant is filled up on egress and internally recycled, it should not drop below 100 unless there is a leak within the coolant system.

Suit Atmosphere:

The atmosphere of the suit is an extremely safety critical system. The partial pressures of each gas in the atmosphere must be kept at the appropriate values. Too much of any one thing can be detrimental. The goal of this system is to keep the partial pressure of oxygen at 4.0 psi, and the partial pressures of everything else at 0.0 psi. There are a lot of factors that can push things out of these bounds.

Once the EVA begins, the astronaut is constantly consuming oxygen and producing carbon dioxide. Oxygen is being supplied from the oxygen tanks that are filled during egress. Carbon dioxide is collected by the carbon dioxide scrubbers and released into space. One of the main issues that early suits found is that the carbon dioxide likes to pool in the helmet and has trouble distributing throughout the suits where it can be more easily collected. To assist with this, fans were added in the helmet to help push the carbon dioxide towards the carbon dioxide scrubbers.

The user should be notified if the suit co2 pressure, the helmet co2 pressure, and/or the suit other pressure are nearing their max nominal value. The user should also be notified if any other suit atmosphere values are nearing the min/max nominal value.

Suit Fans:

The fans distribute carbon dioxide from the helmet to the rest of the space suit. Each suit has two fans, a primary and secondary fan. Only one fan is on at a time. Each fan should be spinning at 30,000 rpm. The speed of the fan determines how much carbon dioxide is moved from the helmet to the rest of the suit. If the fan speed is too low, not enough carbon dioxide will be pushed into the scrubbers and the helmet bubble will fill with carbon dioxide. Too much carbon dioxide in the helmet bubble can impair the astronaut so it must be kept below 0.15 psi.

The user should be notified if the fan speed deviates from 30,000 RPM.

Suit CO2 Scrubbers:

The suit has two alternating carbon dioxide scrubbers. One scrubber is collecting carbon dioxide from the suit while the other is ejecting it's collected carbon dioxide into space. There is a switch on the DCU to alternate the scrubbers. This is necessary because as the EVA goes on, one of the scrubbers will fill up and become less efficient. If this goes unnoticed for too long, the suit will start to fill with carbon dioxide. The astronaut must be told that their scrubber is almost full and that they need to alternate the scrubbers, to start filling up the empty scrubber and ejecting the contents of the full scrubber.

Suit Temperature:

Because of how well the space suit is insulated, the temperature inside the suit fluctuates drastically based on physical activity. There is a record of an astronaut on a lunar EVA being told to slow down because his suit was getting too hot. Our suits will have a simulated temperature value. As the suit gets hot, the coolant will heat up and can turn gaseous while the radiator cools it off. The coolant is filled during the egress process and self manages the temperature throughout the rest of the EVA.

The user should also be warned when their temperature is nearing the maximum nominal value.

Handling Error Scenarios

heart_rate

Alert that the detected heart rate is too high and that the astronaut will need to slow down for a second. `suit_pressure_oxy` The primary oxygen tank is not supplying enough (or too much) oxygen, and we must swap to the secondary oxygen tank. This is done using the oxygen switch on the DCU. Alert the user to return to PR as soon as possible.

suit_pressure_co2

The scrubber has filled up and must be vented. This is done using the carbon dioxide switch on the DCU. `suit_pressure_other` The partial pressure of all other gases should be zero in the suit after the decompress sequence. Alert the user if this value is too high only after that step. Alert the user to return to PR as soon as possible.

suit_pressure_total

The suit total pressure being too low/high alludes to a problem with either the oxygen tank or the scrubber. Review those values and perform their procedures.

helmet_pressure_co2

The helmet carbon dioxide partial pressure builds up on a fan failure. Swap to the secondary fan. This is done using the fan switch on the DCU. Alert the user to return to PR as soon as possible.

fan_pri_rpm and fan_sec_rpm

If either of the fans are on and not spinning at the expected 30,000 rpm, then there is a fan error. Swap to the secondary fan. This is done using the Fan Switch on the DCU. Alert the user to return to PR as soon as possible.

scrubber a co2 storage and scrubber a co2 storage

If either scrubber filled beyond 60% capacity, it must be vented. This will likely happen a few times during the EVA, and the astronaut should be alerted that they need to vent their collected carbon dioxide. This is done by flipping the Carbon Dioxide Switch on the DCU. temperature Alert that the detected temperature is too high and that the astronaut needs to slow down.

primary battery level

If the primary battery level is low (below 20%), switch to secondary battery by switching the battery switch from primary to secondary battery on the DCU.